AI-Based Plant Disease Detection Using Image Classification: Comparative Analysis of Custom CNN and MobileNetV2 for Potato Leaf Disease Classification

Md Monir Hossain — 0222220005101165
Department of Computer Science and Engineering
Premier University, Chattogram, Bangladesh
Email: mdmonirhossain1165@gmail.com

***Abstract*—Plant diseases can significantly affect agricultural productivity, making accurate and timely disease identification important for crop management. This paper presents a comparative evaluation of a Custom Convolutional Neural Network (CNN) and MobileNetV2 for automated potato leaf disease classification. A filtered subset of the PlantVillage dataset containing 2,152 images across three classes—Potato Early Blight, Potato Late Blight, and Potato Healthy—was used. Images were resized to 224×224 pixels, normalized, and divided into training, validation, and testing subsets using a stratified split. The Custom CNN was trained from scratch, while MobileNetV2 used ImageNet-pretrained weights followed by fine-tuning. On the held-out test set, the Custom CNN achieved 94.12% accuracy, 90.45% macro precision, 93.32% macro recall, and 91.78% macro F1-score. MobileNetV2 achieved 97.52% accuracy, 96.87% macro precision, 94.54% macro recall, and 95.64% macro F1-score. MobileNetV2 outperformed the Custom CNN across all four reported aggregate metrics on this test set.**

***Index Terms*—plant disease detection, potato leaf disease, image classification, convolutional neural network, Custom CNN, MobileNetV2, transfer learning, PlantVillage**

I. Introduction

Plant diseases are an important challenge in agricultural production because they can reduce crop quality and yield when they are not identified and managed appropriately. Conventional disease identification commonly relies on visual inspection by farmers or agricultural experts. Although expert inspection can be effective, it can also be time-consuming and may depend on the experience of the observer.

Recent developments in computer vision and deep learning have enabled automated approaches for plant disease recognition from leaf images. Convolutional neural networks (CNNs) can learn visual characteristics such as disease spots, discoloration, texture changes, and other patterns directly from images.

Potato is an important agricultural crop, and diseases such as Early Blight and Late Blight can negatively affect production. Automated classification of potato leaf images can therefore provide a foundation for computer-assisted plant disease identification.

This paper investigates the classification of potato leaf images into three categories: Potato Early Blight, Potato Late Blight, and Potato Healthy. Two approaches are compared: a Custom CNN trained from scratch and MobileNetV2 using transfer learning followed by fine-tuning.

The main objective is to compare the two architectures using the same dataset, preprocessing pipeline, and held-out test set. Accuracy, precision, recall, F1-score, and confusion matrices are used for evaluation.

II. Related Work

Deep learning has become an important approach for automated plant disease recognition because convolutional neural networks can learn discriminative visual features directly from leaf images. Mohanty *et al.* used the PlantVillage dataset to investigate image-based plant disease detection and demonstrated the potential of deep convolutional models for classifying diseased and healthy plant leaves [1].

The availability of large plant-image datasets has further encouraged the development of automated disease classification systems. However, models trained on controlled datasets may encounter difficulties when applied to field images because of variations in lighting, background, camera quality, leaf

orientation, and disease appearance. This creates an important distinction between high benchmark performance and practical field generalization.

Transfer learning provides another approach to plant disease classification. Instead of learning all visual representations from randomly initialized parameters, pretrained convolutional networks can be adapted to a target classification task. MobileNetV2 introduced inverted residual blocks and linear bottlenecks while providing a computationally efficient architecture for image recognition [2]. Its relatively lightweight design makes it particularly relevant for applications where computational resources are limited.

Previous work has demonstrated that pretrained CNN architectures can achieve strong performance in plant disease recognition. At the same time, models trained from scratch remain useful because they provide a direct baseline for understanding the benefit of transfer learning. Therefore, comparing a Custom CNN with a pretrained and fine-tuned architecture under the same experimental conditions provides a useful assessment of the contribution of transfer learning.

In this study, a Custom CNN and MobileNetV2 are evaluated on the same three-class potato leaf classification problem. Unlike a comparison based only on overall accuracy, this study also considers macro precision, macro recall, macro F1-score, class-level performance, and confusion matrices. This provides a more detailed analysis of how the two approaches behave, particularly for the underrepresented Healthy class.

III. Materials and Methods

A. Dataset

The PlantVillage dataset was used as the source of the images. The original dataset contains 54,305 images. A filtered subset containing potato leaf images was prepared for this study.

The final dataset contained 2,152 images belonging to three classes.

The dataset is imbalanced because the Healthy class contains substantially fewer images than the two disease classes.

TABLE I
Distribution of the Potato Leaf Dataset

Class	Images	Percentage
Potato Early Blight	1000	46.47%
Potato Late Blight	1000	46.47%
Potato Healthy	152	7.06%
Total	2152	100%

TABLE II
Dataset Split

Subset	Images	Percentage
Training	1506	70%
Validation	323	15%
Testing	323	15%
Total	2152	100%

B. Data Preprocessing

All images were resized to 224×224 pixels with three RGB channels. Pixel values were normalized using

$$x_{\text{normalized}} = \frac{x}{255}. \tag{1}$$

A stratified split with random seed 42 was used to divide the dataset into training, validation, and testing subsets.

Random horizontal flipping was applied only to training images. The batch size was 32, and TensorFlow's `tf.data` pipeline was used for batching, shuffling, and prefetching.

C. Custom CNN

The Custom CNN was developed from scratch. Its architecture consisted of three convolutional blocks followed by fully connected layers.

The model contained 11,169,347 trainable parameters. Adam was used with a learning rate of 0.001 and sparse categorical cross-entropy loss.

D. MobileNetV2

MobileNetV2 was employed using ImageNet-pretrained weights. The original classification head was removed and replaced with Global Average Pooling, a 128-unit dense layer, dropout, and a three-class Softmax output.

TABLE III
CUSTOM CNN ARCHITECTURE

Layer	Configuration
Input	$224 \times 224 \times 3$
Conv2D	32 filters, 3×3, ReLU
MaxPooling	2×2
Conv2D	64 filters, 3×3, ReLU
MaxPooling	2×2
Conv2D	128 filters, 3×3, ReLU
MaxPooling	2×2
Flatten	–
Dense	128, ReLU
Dropout	0.5
Output	3, Softmax

The complete model contained 4,374,147 parameters.

Initially, the MobileNetV2 base was frozen and the newly added classifier was trained using Adam with a learning rate of 0.001. Subsequently, the last 20 layers of the base network were unfrozen for fine-tuning using a learning rate of 1×10^{-5}.

E. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Accuracy was calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (2)$$

Precision and recall were calculated as

$$Precision = \frac{TP}{TP + FP}, \quad (3)$$

$$Recall = \frac{TP}{TP + FN}. \quad (4)$$

The F1-score was calculated as

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}. \quad (5)$$

Macro-averaged metrics were obtained by taking the arithmetic mean of the corresponding metric across the three classes.

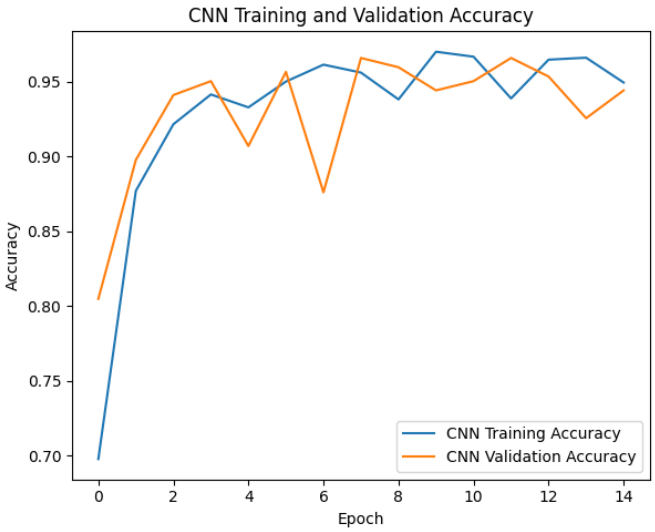


Fig. 1. Training and validation accuracy of the Custom CNN.

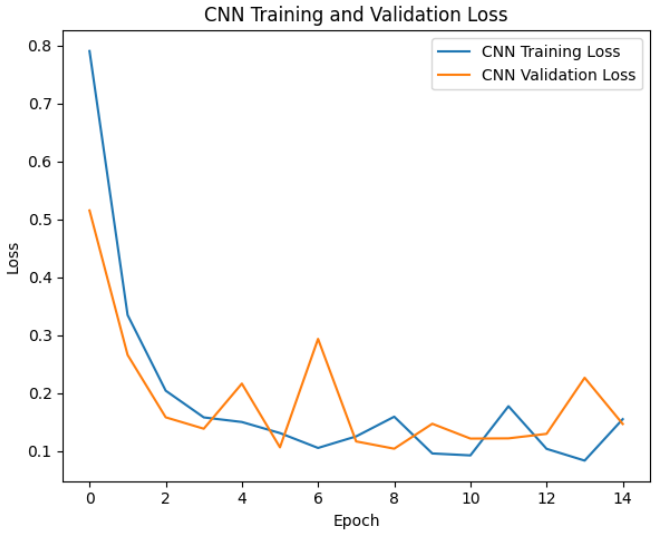


Fig. 2. Training and validation loss of the Custom CNN.

IV. EXPERIMENTAL RESULTS

A. Training Performance

The Custom CNN completed all 15 training epochs. At the final recorded epoch, training accuracy was 98.01% and validation accuracy was 97.52%. The corresponding training and validation losses were 0.0491 and 0.0715.

During frozen-base training, MobileNetV2 achieved 98.61% training accuracy and 98.76% validation accuracy. After fine-tuning, the last recorded valida-

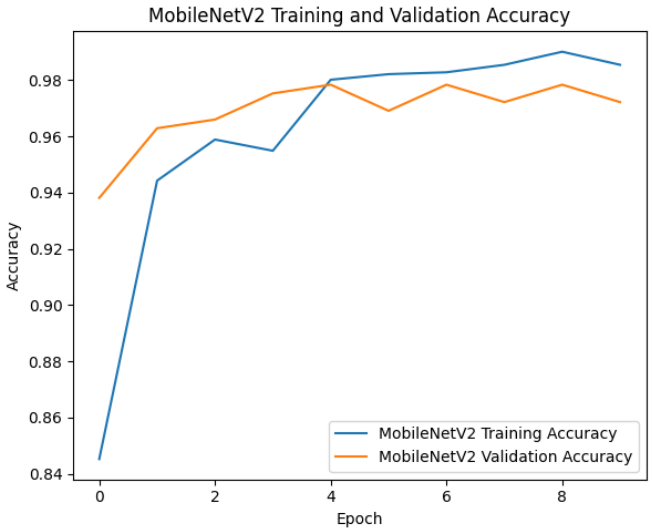


Fig. 3. Training and validation accuracy of MobileNetV2.

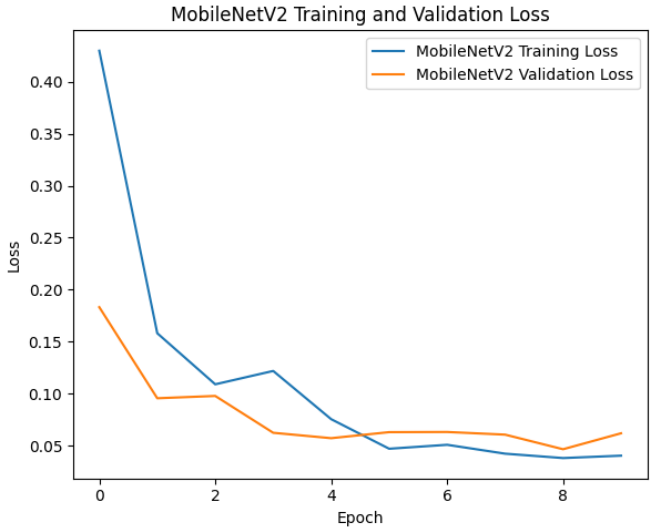


Fig. 4. Training and validation loss of MobileNetV2.

tion accuracy was 99.38%, while training accuracy reached 99.93%.

B. Test Performance

The final comparison was performed on the independent held-out test set containing 323 images. MobileNetV2 achieved 97.52% accuracy compared with 94.12% for the Custom CNN, an improvement of 3.40 percentage points. MobileNetV2 also achieved higher macro precision, macro recall, and macro F1-score in this evaluation.

TABLE IV
OVERALL TEST PERFORMANCE

Metric	Custom CNN	MobileNetV2
Accuracy	94.12%	97.52%
Macro Precision	90.45%	96.87%
Macro Recall	93.32%	94.54%
Macro F1-Score	91.78%	95.64%

TABLE V
CLASS-LEVEL CLASSIFICATION PERFORMANCE

Model	Class	Precision	Recall	F1
Custom CNN	Early Blight	0.9662	0.9533	0.9597
	Late Blight	0.9396	0.9333	0.9365
	Healthy	0.8077	0.9130	0.8571
MobileNetV2	Early Blight	0.9801	0.9867	0.9834
	Late Blight	0.9735	0.9800	0.9767
	Healthy	0.9524	0.8696	0.9091

C. Class-Level Performance

MobileNetV2 achieved higher precision, recall, and F1-score for Early Blight and Late Blight. It also performed better on the Healthy class in terms of precision and F1-score. The Custom CNN achieved higher Healthy-class recall (0.9130 compared with 0.8696), indicating that it identified a larger proportion of the healthy test images.

D. Confusion Matrix Analysis

The Custom CNN confusion matrix was

$$\begin{bmatrix} 143 & 7 & 0 \\ 5 & 140 & 5 \\ 0 & 2 & 21 \end{bmatrix}. \tag{6}$$

The model correctly classified 304 of the 323 test images, giving an accuracy of 94.12%.
The MobileNetV2 confusion matrix was

$$\begin{bmatrix} 148 & 1 & 1 \\ 3 & 147 & 0 \\ 0 & 3 & 20 \end{bmatrix}. \tag{7}$$

MobileNetV2 correctly classified 315 of the 323 test images, giving an accuracy of 97.52%.

V. DISCUSSION

The experimental results show that both architectures achieved strong performance on the selected PlantVillage potato dataset. MobileNetV2 obtained

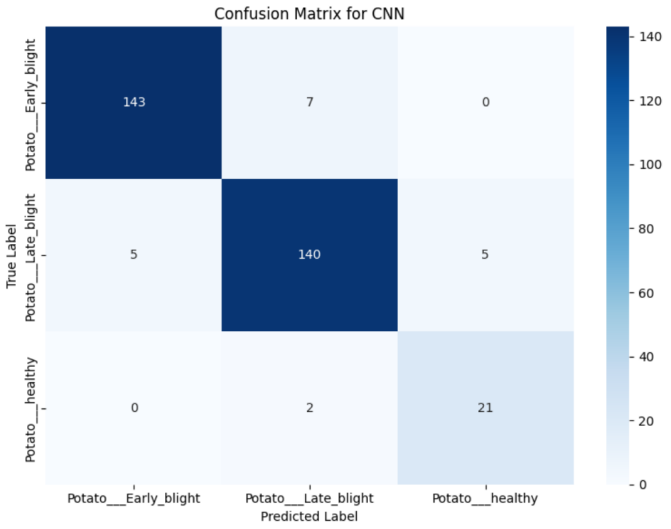


Fig. 5. Confusion matrix of the Custom CNN.

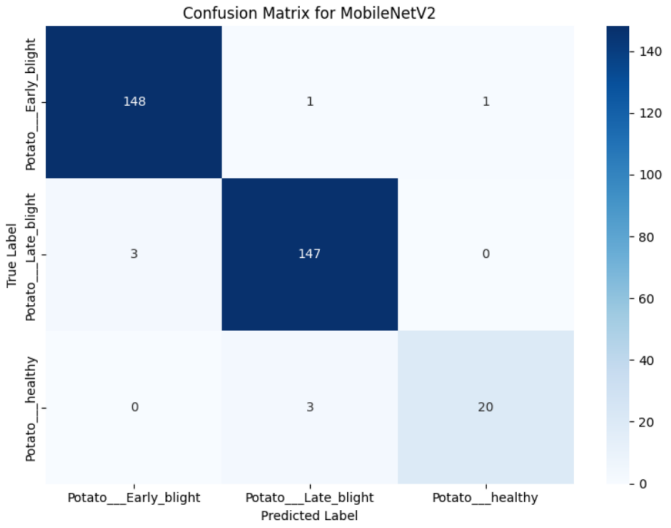


Fig. 6. Confusion matrix of MobileNetV2.

higher overall test accuracy, macro precision, macro recall, and macro F1-score than the Custom CNN on the held-out test set.

The difference in performance is also reflected in the confusion matrices: MobileNetV2 correctly classified 315 of 323 test images, whereas the Custom CNN correctly classified 304.

The Healthy class was particularly important in this comparison because it contained only 152 images in the complete dataset. MobileNetV2 achieved 86.96% recall for this class, whereas the Custom CNN achieved 91.30%. However, MobileNetV2 obtained a higher Healthy-class F1-score (0.9091) than the Custom CNN (0.8571), reflecting a different balance between precision and recall.

The results should be interpreted in the context of the dataset distribution. In particular, the relatively small number of Healthy images means that class-level estimates for this category should be interpreted cautiously.

VI. Limitations

The primary limitation is the imbalance between the three classes. The Healthy class contains considerably fewer samples than the two disease classes.

In addition, the PlantVillage dataset is relatively controlled compared with photographs collected directly from agricultural fields. Differences in lighting, background, camera quality, leaf orientation, and disease severity may therefore affect real-world performance.

The current study also uses a limited set of augmentation techniques and does not include an independent field-collected test dataset.

VII. Future Work

Future work should investigate larger and more balanced datasets, particularly additional Healthy samples. Stronger augmentation, class-weighting, and oversampling techniques could also be evaluated.

Additional pretrained architectures such as EfficientNet, ResNet, and DenseNet could be compared with MobileNetV2. Independent field images should also be used to evaluate generalization beyond the PlantVillage dataset.

Explainable AI methods such as Grad-CAM could further be incorporated to determine which regions of potato leaves contribute to model predictions.

Finally, model compression and quantization could be investigated to support deployment on mobile and edge devices.

VIII. Conclusion

This paper presented a comparative evaluation of a Custom CNN and MobileNetV2 for three-class potato leaf disease classification.

The Custom CNN achieved 94.12% test accuracy, 90.45% macro precision, 93.32% macro recall, and 91.78% macro F1-score. MobileNetV2 achieved 97.52% test accuracy, 96.87% macro precision, 94.54% macro recall, and 95.64% macro F1-score.

MobileNetV2 outperformed the Custom CNN across all four aggregate metrics on the evaluated test set. The Custom CNN nevertheless achieved higher recall for the Healthy class, while MobileNetV2 achieved higher Healthy-class precision and F1-score. These findings demonstrate that both architectures can classify potato leaf images effectively on the evaluated PlantVillage subset.

However, the class imbalance and controlled dataset conditions indicate that further validation using diverse field-collected images is necessary before these models can be considered for practical agricultural deployment.

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